

The background features a large, faint illustration of a jellyfish with a wide, bell-shaped top and long, flowing tentacles. To the right of the jellyfish is a smaller illustration of a fish, possibly a koi, with orange and white stripes. At the bottom of the image, there are dark, jagged shapes representing a rocky or mountainous landscape, likely Mars. The entire scene is set against a warm, orange-to-yellow gradient background.

Saavan'26 IIT MADRAS
ORGANIZED BY AVASAYA SOCIETY

*W*HISPERS FROM MARS

Participant Learning Module

Event	Whispers from Mars · SAAVAN '26
Organiser	AVASYA Society, IIT Madras BS Degree Programme
Purpose	Give every participant the scientific and mission-reasoning foundation needed for the Workshop, Quiz, and case-study rounds
Prerequisite knowledge	None beyond high-school science — this document assumes zero prior astrobiology background.
Companion documents	Whispers from Mars Rulebook (schedule, eligibility, scoring), Martian Packet, and mission brief

How to Use This Document

This module is a foundational reference guide rather than a script to memorize. It explains the core scientific reasoning behind key discoveries so that in Stage 2 (Quiz) you can reason through unfamiliar questions, and in Stages 3–5 (Team Rounds) you can clearly justify your conclusions during live Q&A.

Recommended reading path

If you are preparing for...	Read
Stage 1 (Workshop)	Modules 1 → 4, in order, once
Stage 2 (Quiz)	Modules 1 → 4 in full, plus the Glossary for quick term checks. Pay particular attention to Module 3.4 (the reasoning checklist) and Module 3.5 (the CoLD scale).
Stage 3 (First Signal)	Module 3, section 3.4 (reasoning checklist), before anything else
Stage 4 (Mission Brief)	Module 4 in full, as a team — plus Appendix C if your team wants to work with real PIXL data
Stage 5 (Mission Defence)	Module 4.3 and Module 3.5, to rehearse your defence in the language judges will actually use

Table of Contents

1. Orientation — The Story in 90 Seconds
2. Module 1 — The Habitability Triad
3. Module 2 — Reading a Martian Rock: Instruments & Data
4. Module 3 — The Trap of False Positives (incl. the CoLD Scale)

5. Module 4 — The Engineering & Mission-Director Crisis
6. Glossary
7. Appendix A — Source Material Map
8. Appendix B — Who to Ask
9. Appendix C — Go Deeper: Verified NASA & Data Resources

1. Orientation — The Story in 90 Seconds

In July 2024, NASA's Perseverance rover explored an ancient river channel called Neretva Vallis inside Mars's Jezero Crater. There, it found an arrowhead-shaped, rusty-red mudstone rock nicknamed Cheyava Falls. Its rusty-red color comes from iron oxides, proving that liquid water once flowed through the area and altered the rock. The rock is covered in tiny, millimeter-scale "leopard spots"—pale inner areas surrounded by dark outer rings—along with small "poppy seed" textures.

Using its onboard instruments (PIXL and SHERLOC), Perseverance confirmed that Cheyava Falls contains organic carbon alongside sulfur, iron, and phosphorus. These leopard spots offer an intriguing clue to ancient microscopic respiration: on Earth, microbes harvest energy through anaerobic respiration without oxygen by using organic carbon and transferring electrons via iron and sulfur. This metabolic process leaves behind distinct mineral rings—dark outer rings made of vivianite (an iron phosphate) and pale inner centers made of greigite (an iron sulfide). Perseverance drilled a core sample from this rock, sealed it inside Sample Tube 25, and cataloged it as "Sapphire Canyon."

Scientific classification and abiotic mimics

Despite these exciting clues, NASA officially classifies Cheyava Falls as a "potential biosignature" rather than proof of past life. Non-biological (abiotic) processes—such as extreme heat, highly acidic conditions, or low-temperature chemical reactions—can mimic these exact mineral patterns. While surrounding rocks show no signs of past high heat or strong acid, scientists must rigorously test non-biological explanations before claiming a discovery of life. [Hurowitz, J.A. et al., "Redox-driven mineral and organic associations in Jezero Crater Mars"](#)

Bringing Sapphire Canyon back to Earth for laboratory analysis faces huge obstacles. Mars Sample Return (MSR) requires eight unprecedented, high-risk steps in sequence: (1) landing a massive vehicle precisely on Mars, (2) transferring sample tubes robotically between vehicles, (3) launching a rocket (the Mars Ascent Vehicle, or MAV) off another planet for the first time, (4) performing autonomous rendezvous and capture in Mars orbit, (5) robotically sealing the sample container, (6) departing Mars orbit for Earth, (7) capturing the return capsule in Earth orbit, and (8) safely landing extraterrestrial material under strict containment standards.

The MAV presents an immense engineering challenge: it must survive freezing Martian nights stored sideways before attempting its single, un-testable launch attempt with zero option for repairs. Consequently, overall MSR cost estimates grew to \$8–11 billion. In early 2026, the U.S. Congress defunded the program as originally designed, and ESA retreated from its planned partnership. Meanwhile, geopolitical competition is rising:

China's Tianwen-3 mission is actively preparing its own Mars sample return from a different site, meaning independently collected Martian samples could reach an Earth lab first.

The Central Question

Your goal across all stages of this event is to think like a mission scientist: Is the current evidence proof of life, non-biological activity, or genuine uncertainty? Given limits on budget, time, and instruments, what should NASA do next? Where does today's evidence sit on NASA's confidence ladder (Module 3.5) — and what specific data would move it one step higher?

Must-Watch / Highly Recommended Resource: For a comprehensive video overview covering key details discussed in this module, watch this video: [Whispers from Mars - Recommended Overview \(YouTube\)](#)

Field Note — Percy Wasn't Working Alone

For nearly three years, a second spacecraft flew scouting missions over exactly this terrain: Ingenuity, a 1.8 kg helicopter that hitched a ride on Perseverance's belly and made the first powered, controlled flight on another planet. Built to attempt five short hops, it ended up flying dozens of times — scouting the delta and canyon terrain ahead of the rover — before rotor-blade damage on its final landing grounded it for good.

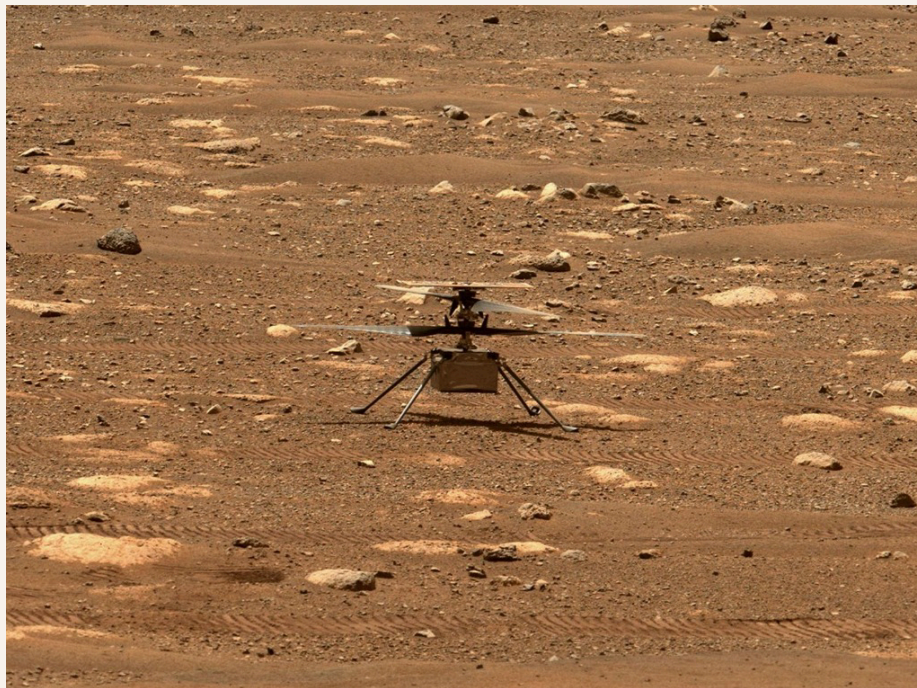


Figure — Ingenuity on the Martian surface

Ingenuity sitting on the Martian regolith between flights, rotors at rest. It never carried scientific instruments — its entire job was to prove aerial scouting was possible, and to show mission planners where Percy should drive next.

2. Module 1 — The Habitability Triad

Every claim of past or present life on Mars rests on three independent pillars. If any one is missing, the case is weak — this is the first filter to apply to any evidence packet.

The Habitability Triad

LIQUID WATER + CHNOPS + AN ENERGY SOURCE

(the solvent)

(the building blocks)

(the reason to bother)

1.1 Liquid Water — the solvent

Mars's current atmospheric pressure is so low that liquid water vaporises almost instantly on the surface; it survives only underground or under ice today. Billions of years ago, before Mars lost its magnetic field and its atmosphere thinned, water flowed freely.

Why Jezero Crater: Satellite imagery reveals a prominent ancient river delta fanning into Jezero Crater alongside an outflow channel, proving a standing lake was sustained here over an extended geological period.

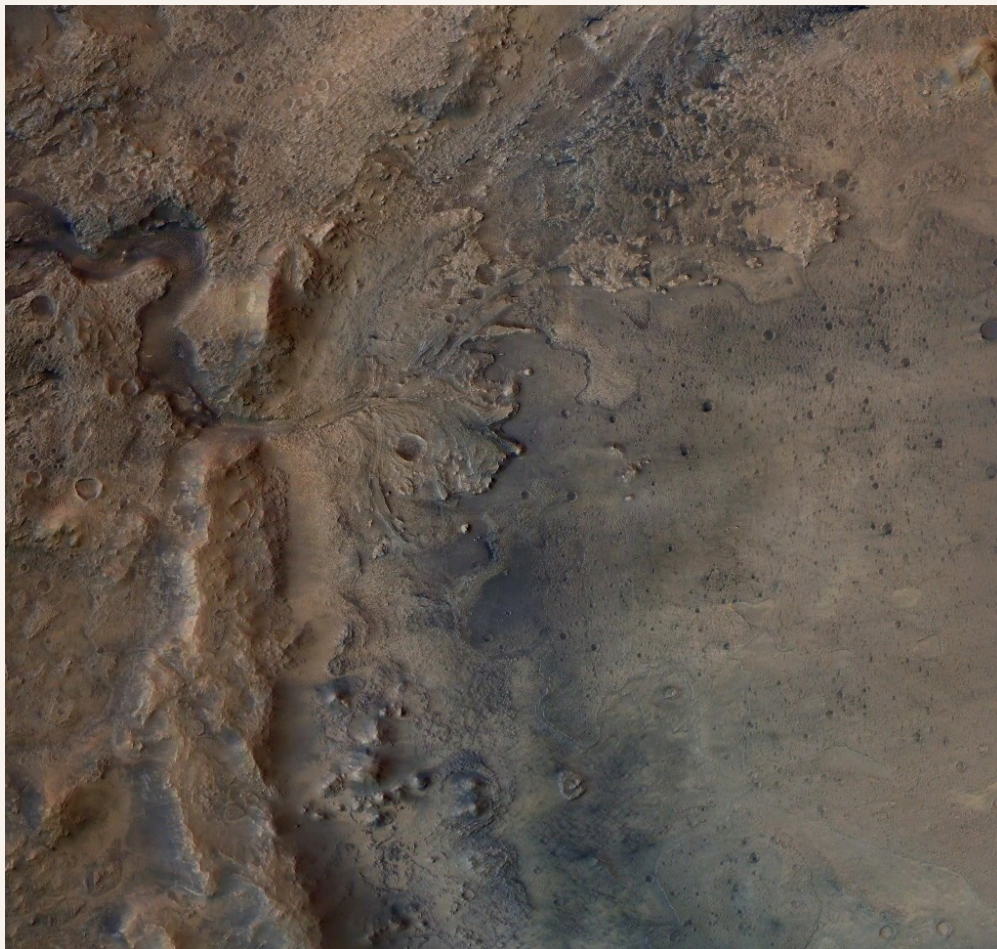


Figure 1 — The Jezero Crater delta, imaged from Mars orbit

The fan-shaped fossil delta (upper-left) spilling into the crater floor is the single feature that made Jezero NASA's top pick for Perseverance's landing site. Where the channel meets open water, the current slows and drops its sediment — which is exactly the reasoning in the box below.

Key Reasoning

Why deltas matter: Fine mud settling slowly in quiet water is exceptional at preserving ancient biological and organic signatures. Fast-flowing water breaks delicate structures apart, whereas calm sediment traps and buries them safely. Deltas are top astrobiology targets because of this preservation potential — not because of landing safety or present-day surface water.

1.2 CHNOPS — the building blocks

Life as we know it is built overwhelmingly from six elements:

Letter	Element	Role
C	Carbon	Backbone of organic molecules
H	Hydrogen	Bonds, water, energy carriers
N	Nitrogen	Proteins, DNA/RNA bases
O	Oxygen	Water, respiration, oxides
P	Phosphorus	DNA/RNA backbone, energy storage (ATP)
S	Sulfur	Proteins, some metabolisms

Finding CHNOPS elements together is necessary, not sufficient — it establishes that life could work with the available raw material, nothing more.

1.3 Energy — “life is an electron looking for a place to rest”

This is the single most important idea in the event.

Core mechanism: living things generate energy by moving electrons from something with electrons to spare (a donor) to something that wants to receive them (an acceptor). As the electron moves, the cell captures some of the energy released — comparable to a ball rolling downhill and turning a turbine on the way down.

System	Donor	Acceptor
Humans	Carbon-rich food	Oxygen (why we breathe)
Rock-eating microbes (Earth)	Minerals in rock	Iron or sulfate — anaerobic respiration

General term for “getting energy from inorganic rock chemistry, without sunlight”: **chemolithotrophy**.

Why this matters for Mars: in 2013, Curiosity found an electron donor and acceptor together in an ancient Martian lakebed — described by mission scientists as finding a battery whose contents were never confirmed

as used. That confirmed habitability. Cheyava Falls potentially shows the chemical residue of a battery that was used — a step beyond habitable toward inhabited.

Redox, in one line

- Oxidation = losing electrons
- Reduction = gaining electrons (mnemonic: “RIG” — Reduction Is Gain)
- On Cheyava Falls, iron(III) gaining an electron to become iron(II) is a reduction reaction — part of what forms vivianite and greigite.

3. Module 2 — Reading a Martian Rock: Instruments & Data

2.1 Perseverance's key instruments

Instrument	What it does	What it cannot do
PIXL (Planetary Instrument for X-ray Lithochemistry) — PI: Abigail Allwood, JPL	Fires a focused X-ray beam at a spot as small as a grain of salt and maps which elements are present, and their concentration, pixel by pixel	Identify specific minerals directly (it maps chemistry, not mineralogy); image anything at cellular scale
SHERLOC (Scanning Habitable Environments with Raman & Luminescence for Organics & Chemicals) — PI: Luther Beegle, JPL	Uses a UV laser (Raman/fluorescence) to detect minerals and organic compounds	Map bulk elemental composition the way PIXL does
Mastcam-Z	High-definition stereoscopic and panoramic imaging system mounted on the rover mast	Provide atomic or chemical composition data directly
SuperCam	Laser-induced spark analysis (LIBS) plus Raman spectroscopy, from a distance	Match the resolution of up-close instruments
WATSON (Wide Angle Topographic Sensor for Operations and eXploration) — integrated with SHERLOC	High-resolution close-up camera providing visual context and micro-texture analysis alongside SHERLOC	Perform chemical analysis on its own (relies on SHERLOC for spectroscopy)
MOXIE (Mars Oxygen ISRU Experiment)	Demonstrates technology to produce pure oxygen from Martian atmospheric CO ₂ for future human missions	Detect biosignatures or analyze rock mineralogy

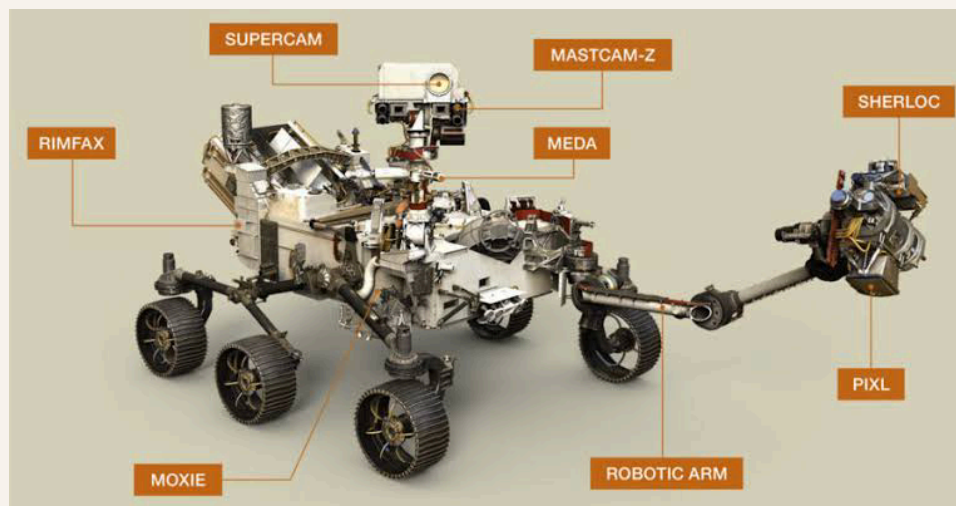


Figure 2 — Perseverance's instrument suite

Four more instruments ride on the rover beyond the two above. Mastcam-Z and SuperCam sit on the mast for long-range imaging and remote spectroscopy; MEDA is the onboard weather station (pressure, wind, temperature, dust); RIMFAX is ground-penetrating radar, mapping subsurface layers the cameras can't see; MOXIE (retired 2023) demonstrated making oxygen from Martian CO₂ as a test for future human missions. PIXL and SHERLOC — the two instruments Module 2 focuses on — sit together on the turret at the end of the robotic arm, so they can be placed directly against a rock.

Distinction to keep straight: PIXL maps elements (chemistry); SHERLOC/WATSON identify minerals, organic matter, and texture. They are complementary, not redundant. [Mars 2020 Fact Sheet, NASA](#)

Engineer's Aside — What PIXL Actually Is

PIXL is a microfocus X-ray fluorescence (XRF) spectrometer. It fires X-rays at the rock; each element's atoms absorb the energy, kick out an inner-shell electron, and re-emit a characteristic X-ray as an outer electron falls in to fill the gap. The energy of that re-emitted photon identifies the element; the number of photons at that energy (the counts) relates to its concentration.

Every reading from PIXL is a statistical measurement. Low-count spectral peaks carry more statistical noise than strong signals, so subtle chemical detections require far greater scrutiny before drawing major scientific conclusions.

2.2 Reading an elemental map

PIXL output is a false-colour map, not a photograph — and the colour assignment is chosen by whoever processes the data for a given figure, not a fixed universal code. So the first thing to do with any map you're handed is read its legend, not assume a colour convention. What is fixed is the method:

1. Where are elements concentrated — centre, ring, or uniform?
2. Which elements co-occur in the same location — that indicates the mineral forming there (e.g., iron + phosphorus co-occurring points toward an iron-phosphate mineral such as vivianite)
3. Does the spatial pattern match a known chemical process, or does it look like noise? On Cheyava Falls specifically, this method is exactly what identified vivianite (iron + phosphorus, in the dark rims and “poppy

seeds”) and greigite (iron + sulfur, in the pale interiors) as the minerals forming the “leopard spot” reaction fronts — see Module 3.1.

2.3 Defining “biosignature” precisely

A biosignature is any measurable substance, pattern, or structure whose formation is best explained by biological activity. It is a suspect, not a verdict — it raises the probability of life without proving it. [“What Is a Potential Biosignature?” — NASA](#)

Real Data You Can Explore

If your team reaches Stage 4, these are the actual tools and archives real Mars scientists use — not simulations:

- [PIXLISE](#) — the open-source visual-analytics platform scientists use to explore PIXL's X-ray maps (built by JPL and Queensland University of Technology).
- [Mars 2020 Analyst's Notebook](#) — mission data, instrument readings, and documentation, maintained by the PDS Geosciences Node at Washington University in St. Louis.
- [Mars Rock Samples catalogue](#) — NASA's page on every core collected so far, including Sapphire Canyon.

Always cross-check observational data against independent spectral lines before forming conclusions.

4. Module 3 — The Trap of False Positives

This module underlies the scoring rubric for Stages 3–5. Teams lose the most points here.

3.1 Biotic vs. abiotic — always ask “what else could do this?”

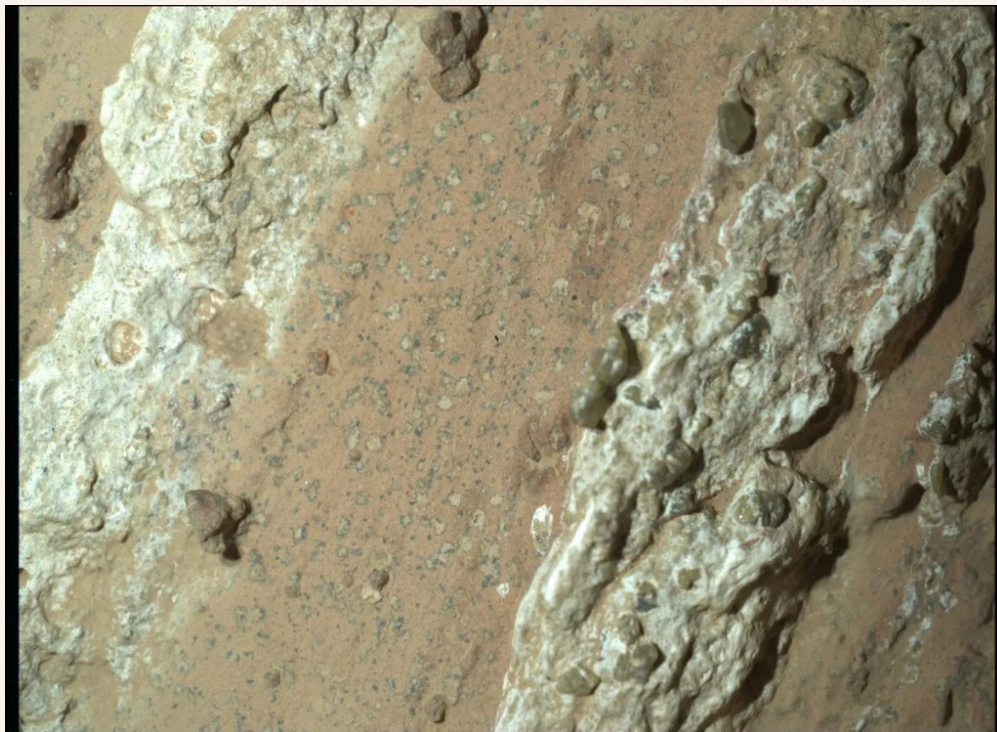


Figure 3 — Cheyava Falls's "leopard spots," up close

A macro view of the reaction fronts: darker rims (vivianite, an iron phosphate) encircling paler, off-white interiors (greigite, an iron sulfide). The two smaller pale nodules with dark speckling to the left are "poppy seeds" — the same vivianite chemistry on a smaller scale.

Biological pathway: as microbes drew electrons from nearby iron to harvest energy, that iron shifted from rust-red iron(III) to pale iron(II) — precipitating vivianite (an iron phosphate) at the reaction front, visible today as the dark rim of each leopard spot and in the smaller "poppy seed" nodules. Deeper in, where the reaction ran to completion and the accessible iron was used up, microbes would have switched to sulfate as an electron acceptor instead — precipitating greigite (an iron sulfide) in the resulting pale core.

Abiotic pathway — the mimic: the same two minerals can form without any biology, through sustained high temperature, acidic conditions, or organic-compound catalysis at low temperature. This is the specific trap built into this event: thermal, acidic, or catalytic mimicry of iron-phosphate and iron-sulfide minerals.

Cheyava Falls remains compelling because surrounding rocks show no signs of past high heat or extreme acidity. However, whether ambient organic compounds could have driven these reactions without biological assistance remains open. That is why NASA formally designates the finding as a "potential biosignature" rather than definitive evidence of past life. [NASA Announcement, September 2025](#)

3.2 The cautionary precedent: the 1996 Mars meteorite

In 1996, researchers announced possible microscopic fossils inside a Martian meteorite (ALH84001). Subsequent studies demonstrated that non-biological geological processes could form identical shapes, leading the broader scientific community to reject the biological claim. This famous precedent explains NASA's rigorous caution today: extraordinary claims demand exhaustive verification.

3.3 What would actually resolve the question

On-Mars instruments can only produce suggestive, never definitive, evidence, because ruling out every abiotic mimic requires building-sized, nanoscale-precision lab equipment. Bringing Sapphire Canyon to Earth would allow:

- Iron isotope analysis — biological processing shifts iron isotope ratios in a recognisable way distinct from non-biological processing
- Sulfur isotope analysis — same principle, applied to the sulfur in the halos versus the surrounding mud
- Carbon-chain length analysis — biological membrane chemistry strongly favours even-numbered carbon chains; abiotic organic chemistry shows no such preference

Exam-ready distinction: the bottleneck is not “more time” or “sharper cameras” — it is that isotope-ratio and molecular-chain analysis require lab-grade instruments that cannot be miniaturised to flight scale.

3.4 Reasoning checklist — apply this to any evidence packet

1. What does the evidence show? (observation only, no interpretation)
2. What is the most exciting interpretation? (the “life” story)
3. What is at least one non-biological interpretation that fits equally well?
4. What specific new measurement would separate the two?
5. What does the evidence not tell us at all?
6. Given all this, what should happen next — investigate further, declare sufficient, or stop?

Scoring Note

Per the Rulebook: submissions are evaluated on evidence-based reasoning, identification of alternative explanations, recognition of uncertainty, quality of proposed tests, and separation of observation from conclusion. A well-justified “uncertain” scores higher than an overconfident “life,” and higher than an overconfident “not life.”

3.5 The CoLD Scale — NASA's Own Confidence Ladder

In 2021, a team led by NASA's then-chief scientist James Green proposed the Confidence of Life Detection (CoLD) scale — a 7-level framework for how confident the scientific community should be that a given observation is evidence of life beyond Earth. It was deliberately modelled on a scale most engineering students already know: NASA's own Technology Readiness Level (TRL) system (1–9), used across the agency to grade how mature a piece of hardware is before it's trusted on a real mission. CoLD does the same job for a scientific claim instead of a technology. [Technology Readiness Levels — NASA](#) , [The Planetary Society](#). ["Confidence of Life Detection Scale"](#)

Level	What has to be established
1	A potential biosignature is detected — a chemical, structural, or textural signal consistent with a biological origin
2	Contamination from Earth (on the instrument, the spacecraft, or the sample chain) is ruled out

Level	What has to be established
3	The environment is shown to be capable of actually producing or hosting the signal — habitability is demonstrated, not just plausible
4	A biological explanation is shown to be the best among the abiotic alternatives proposed at the time of discovery
5	An independent line of evidence — a different instrument or method — corroborates the original signal
6	New alternative (abiotic) hypotheses proposed by the wider community in response to the finding are also ruled out
7	A further, independent signal predicted by the “life” hypothesis is specifically looked for and found — the closest the framework gets to confirmation

Where does Cheyava Falls / Sapphire Canyon sit on the CoLD scale? Currently, it sits at the early rungs (Levels 1–4). While scientists have detected exciting suggestive signs that have passed peer review, non-biological (abiotic) explanations cannot yet be fully ruled out on Mars. To reach the higher levels of proof (Levels 5–7), the sample must be brought back to an Earth laboratory for advanced testing, which is why the Mars Sample Return mission is essential to solving this scientific question.

5. Module 4 — The Engineering & Mission-Director Crisis

Stages 4–5 shift you from scientist to NASA Mission Director, where science meets budget, risk, and politics.

4.1 Why the sample can't simply be brought home

Mars Sample Return (MSR), as originally designed, required eight actions that had never been performed, in sequence:

1. Land the largest vehicle ever sent to Mars, at a highly precise location
2. Transfer samples robotically from one Mars vehicle to another
3. Launch a rocket from another planet for the first time (the Mars Ascent Vehicle, MAV)
4. Perform the first autonomous rendezvous and capture in Mars orbit
5. Robotically seal the container to prevent Martian material contamination
6. Depart Mars orbit for the return to Earth
7. Capture the returning container in space
8. Perform a controlled, contained entry and landing of extraterrestrial material on Earth

Any single failure loses the sample permanently — there is no repair capability on Mars. In TRL terms (Module 3.5), the Mars Ascent Vehicle in particular sat at a low technology readiness level: no rocket had ever launched from another planet's surface, so there was no way to reach TRL 9 (“proven in mission operations”) without actually flying it — a live example of why engineers distinguish “we’ve tested this” from “we’ve tested something like this.”

4.2 Where the mission stands now

Factor	Detail
Cost growth	Independent review estimates put the total cost at roughly \$8–11 billion — on top of billions already spent on Perseverance itself — putting MSR in the same budget class as the James Webb Space Telescope
FY2026 funding decision	U.S. Congressional funding for the baseline MSR architecture was reduced to \$0 in FY2026, prompting NASA to seek alternative, lower-cost mission concepts.
ESA's position	The European Space Agency formally moved to step back from its part of the MSR partnership in March 2026, and has discussed repurposing its planned Earth Return Orbiter for other Mars science instead
What survives	A “Mars Future Missions” technology-development line continues at roughly \$110 million/year — funding groundwork (e.g., landing-system technology) that could seed a future, cheaper sample-return concept
The hardest engineering piece	The Mars Ascent Vehicle: it had to survive months stored sideways through frigid nights (careful thermal/fuel control required), then perform flawlessly on its only launch attempt, with no test run possible under real Mars conditions and no repair option
Planetary-protection burden	A non-zero possibility of native Martian microbial life means the sample must be handled to biosafety-lab-grade containment standards, adding major engineering and cost complexity
Geopolitical context	China's Tianwen-3 mission is in active development toward its own Mars sample-return attempt from its own separate landing site — it does not retrieve Perseverance's cached tubes, but it could plausibly get independently collected Martian material into a lab before any NASA/ESA mission does

Discussion point your team should be ready for: the historic Cheyava Falls finding may end up being corroborated — or challenged — using a completely different rock, collected by a completely different country's mission, before Sapphire Canyon itself ever reaches a lab. NASA holds the most studied sample; it currently lacks a funded mission to retrieve it.

Engineer's Lens

Evaluating MSR requires an engineering trade study: balancing past investments against remaining costs, technical risks, and budget constraints. As a mission director, you must make pragmatic, well-reasoned choices under strict real-world limits.

4.3 Structuring a Mission Director's decision

Every Stage 4–5 deliverable maps onto three components, which mirror the judging rubric directly:

Component	What judges are checking
1. Verdict (LIFE / NOT LIFE / UNCERTAIN)	Does the reasoning explicitly weigh supporting evidence against confidence-reducing gaps and named alternatives? Where would your team place the evidence on the CoLD scale (3.5), and why? “Uncertain,” well-argued, is not a weaker answer
2. Mission plan	Is the proposed instrument/measurement specific and realistically obtainable within engineering, time, and budget constraints — not a vague “do more tests”?

Component	What judges are checking
3. Risk analysis	One accepted risk and one rejected risk, each with explicit reasoning — judges are assessing your reasoning, not expecting a risk-free plan (none exists)

6. Glossary

Term	Definition
Biosignature	Any pattern whose formation is best explained (not proven) by biology
Potential biosignature	NASA's precise term for a substance or structure that might have a biological origin but requires more data before a conclusion can be reached — the term used for Sapphire Canyon
Abiotic	Non-living; not produced by biological activity
CHNOPS	Carbon, Hydrogen, Nitrogen, Oxygen, Phosphorus, Sulfur — life's core elements
Redox reaction	A chemical reaction involving electron transfer (oxidation = electron loss; reduction = electron gain)
Chemolithotrophy	Obtaining energy from inorganic chemical reactions in rock/minerals, without sunlight
Anaerobic respiration	Obtaining energy without oxygen as the final electron acceptor
Vivianite	A hydrated iron phosphate mineral; identified in the dark rims and “poppy seeds” of Cheyava Falls's “leopard spots” (Hurowitz et al. 2025)
Greigite	An iron sulfide mineral; identified in the pale cores of Cheyava Falls's “leopard spots” (Hurowitz et al. 2025)
Leopard spots / poppy seeds	Field nicknames for the millimetre-scale reaction-front textures and nodules found on Cheyava Falls
Hydrothermal alteration	Rock chemically altered by hot water flowing through it after formation — a common abiotic mimic of biosignatures
PIXL	X-ray fluorescence instrument mapping elemental composition on Perseverance
SHERLOC	UV Raman instrument detecting minerals and organic compounds
CoLD scale	NASA's 7-level Confidence of Life Detection framework for grading how strong a life-detection claim is, modelled on the TRL scale
TRL (Technology Readiness Level)	NASA's 1–9 scale for how mature a piece of hardware or technology is before it can be trusted on a real mission
MAV	Mars Ascent Vehicle — the rocket intended to launch samples off Mars; never yet flown
MSR	Mars Sample Return — the mission (in its originally designed form) defunded by the U.S. Congress in January 2026
Sapphire Canyon	The official name of the rock core (from Cheyava Falls) sealed in Sample Tube 25

Term	Definition
Tianwen-3	China's own Mars sample-return mission, in active development, targeting a separate landing site from Perseverance
Isotopic analysis	Comparing ratios of different atomic-weight forms of an element to distinguish biological from non-biological processing
False positive	A non-living process that convincingly mimics a sign of life

7. Appendix C — Go Deeper: Verified NASA & Data Resources

Every link below was checked against the live source while preparing this edition. If you're short on time, read the first three; if your team reaches Stage 4, the data-tool links let you work with the same platforms real Perseverance scientists use.

Resource	What it's for	Link
NASA Press Release: NASA Says Mars Rover Discovered Potential Biosignature	Official NASA press release announcing the peer-reviewed Cheyava Falls findings	nasa.gov ↗
NASA Science Mission Hub: Perseverance Rover Overview & Cheyava Falls Feature	NASA's official mission hub covering the Cheyava Falls discovery and mission background	science.nasa.gov ↗
NASA Science Guide: What Is a Potential Biosignature?	Official NASA resource defining potential biosignatures and evaluation criteria	science.nasa.gov ↗
Hurowitz et al. 2025, Nature	The peer-reviewed paper itself: "Redox-driven mineral and organic associations in Jezero Crater, Mars"	nature.com ↗
Whispers from Mars Recommended Overview	Comprehensive video overview covering key details discussed in this module	youtube.com ↗
Mars Rock Samples Catalogue	NASA's page on every core collected so far, including Sapphire Canyon	science.nasa.gov ↗